

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0508

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

Dr.G.Ayyappan

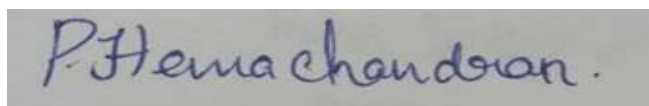
QUESTIONS			Total Marks: 10
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	1
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	1
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	1
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	1
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	1
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	1
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	1

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	1
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	1
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	1

Code of Conduct:

I P. HEMACHANDRAN (192525440) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

Time Managem ent	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretat ion	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Which of the following is true about 1NF?
 - a. Allows multi-valued attributes
 - b. Eliminates partial dependencies
 - c. All attributes must be atomic
 - d. Eliminates transitive dependencies

e. Answer (c) All attributes must be atomic
2. A relation is in 2NF if it is in 1NF and:
 - a. Has no transitive dependencies
 - b. Has no partial dependencies on primary key
 - c. Every attribute is a prime attribute
 - d. Has no multi-valued dependencies

e. Answer (b) Has no partial dependencies on primary key
3. Which normal form deals with multi-valued dependencies?
 - a. 2NF
 - b. 3NF
 - c. BCNF
 - d. 4NF

e. Answer (d) 4NF
4. BCNF is a stricter version of:
 - a. 1NF
 - b. 2NF
 - c. 3NF
 - d. 4NF

e. Answer (c) 3NF
5. Functional dependency $X \rightarrow Y$ means:
 - a. Y determines X
 - b. X determines Y
 - c. X and Y are unrelated
 - d. Both determine each other

e. Answer (b) X determines Y
- 6.
7. Which of the following anomaly is NOT associated with unnormalized relations?

- a. Insertion anomaly
- b. Deletion anomaly
- c. Update anomaly
- d. Retrieval anomaly
- e. Answer (d) Retrieval anomaly

7 . A superkey is:

- a. A minimal set of attributes that uniquely identifies tuples
- b. Any set of attributes that uniquely identifies tuples
- c. An attribute that depends on a non-prime attribute
- d. A foreign key reference

Answer (b) Any set of attributes that uniquely identifies tuples

8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- a. Partially dependent on A
- b. Fully dependent on A
- c. Transitively dependent on A
- d. Independently determined
- e. Answer (c) Transitively dependent on A

9. 5NF is also known as:

- a. Domain-Key Normal Form
- b. Boyce-Codd Normal Form
- c. Project-Join Normal Form
- d. Element Normal Form
- e. Answer (c) Project-Join Normal Form

10. Lossless-join decomposition ensures:

- a. No data is lost during decomposition
- b. All FDs are preserved
- c. The relation is in BCNF
- d. No null values exist
- e. Answer (a) No data is lost during decomposition